

Advancing irrigation strategies for wheat production in Morocco: Combining AquaCrop model and optimization techniques

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ABSTRACT

Global water scarcity poses a significant challenge to agricultural sustainability, particularly in semi-arid regions like Morocco. This study evaluates the benefits of coupling AquaCrop-OSPy with optimization algorithms to improve irrigation scheduling for wheat, aiming to reduce water use while maintaining yield and enhancing water productivity. The optimization framework integrates two methods within AquaCrop-OSPy: (i) a unidimensional optimization based on the *minimize_scalar* bounded algorithm and (ii) a multidimensional optimization using the L-BFGS-B algorithm. This coupling enables the automated adjustment of irrigation thresholds within the model, allowing efficient identification of water-saving yet yield-preserving irrigation schedules. Experiments were conducted over two growing seasons (2016/2017 and 2017/2018) in two wheat fields in the Haouz Plain, Morocco: Field 1 (stressed conditions) and Field 2 (normal conditions). Model calibration using canopy cover, biomass, and actual evapotranspiration (ETa) achieved R² values between 74% and 93%, RMSE values of 0.09–0.10 for CC, 0.15–0.44 t/ha for biomass, and 0.52–0.61 mm/day for ETa. The NSE ranged from 0.71 to 0.95, and the Index of agreement (d) ranged from 0.86 to 0.98. Validation in the second season confirmed similar performance, demonstrating the reliability of AquaCrop-OSPy for wheat simulation under semi-arid conditions. Three irrigation strategies were evaluated: Real Irrigation (farmers' practice), NET irrigation, and Soil Moisture Target (SMT). NET irrigation applies a constant soil-water threshold throughout the crop cycle, whereas SMT uses stage-specific thresholds aligned with key phenological phases (emergence, canopy expansion, maximum canopy, and senescence). Optimization produced substantial improvements. In Field 1, optimized SMT reduced irrigation by 26–31% relative to farmers' practices and 23–31% relative to the pre-optimized SMT baseline, while sustaining yields of 6.43 t/ha in S1 and 6.24 t/ha in S2 and increasing $WP_{Irrigation}$ to 2.08–2.33 kg/m³. In Field 2, optimized SMT reduced irrigation by 42–45% compared to farmers (23–31% relative to pre-optimization), while maintaining yields of 6.43 t/ha in S1 and 6.24 t/ha in S2, with similarly high water productivity. Optimized NET achieved 4–34% savings relative to farmers (10–11% relative to pre-optimization), with moderate gains in $WP_{Irrigation}$. These results demonstrate the clear advantage of SMT over NET in achieving a more favorable water–yield balance. This study confirms that coupling AquaCrop-OSPy with optimization algorithms can substantially improve irrigation scheduling and WP, providing a practical tool for sustainable crop management in water-scarce regions.

1. Introduction

Water scarcity poses major challenges to the sustainability of agriculture, food security, and economic stability in arid and semi-arid

regions. In Morocco, this issue is critical, as agriculture consumes over 70% of the country's freshwater resources (Provenzano and Sinobas, 2014). The sector's heavy dependence on rainfall, particularly for staple crops such as wheat, makes it highly vulnerable to climate variability

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and growing population pressure (Schilling et al., 2012). The imbalance between water demand and supply further emphasizes the need for efficient irrigation strategies. Climate change is compounding Morocco's existing water scarcity challenges (Eddamiri et al., 2024; Hakam et al., 2024). Rising temperatures, irregular rainfall, and more frequent droughts are reducing both surface and groundwater availability, making water supply less predictable (Hssaisoune et al., 2020; Sabri et al., 2024). Recurrent droughts, particularly in regions such as the Tensift watershed have significantly reduced water inflows to the Takerkoust dam, severely limiting irrigation capacity and in some extreme years, leading farmers to entirely abandon annual crops, resulting in a complete loss of production on affected surfaces (Meliho et al., 2019; Bousbaa et al., 2022, 2024; Abou Ali et al., 2023; Nifa et al., 2023; Elyoussfi et al., 2025). Recent official statistics indicate that, despite a sown area of about 2.4–2.6 million hectares of soft wheat, durum wheat and barley, national cereal production has declined by more than 40% in some recent drought-affected campaigns, underlining the severity of Morocco's current water crisis. With projections indicating continued declines in water availability, it is imperative to adopt adaptive agricultural practices that enhance climate resilience.

In the Haouz Plain, these pressures translate into particularly acute constraints for irrigated wheat. Annual rainfall typically ranges around 240–250 mm, while reference evapotranspiration (ET₀) can reach about 1600 mm per year, creating a large atmospheric demand that must be primarily met by irrigation (Chehbouni et al., 2008; Kaissi et al., 2023, 2025a). The plain covers roughly 6000 km², of which about 3000–3100 km² are irrigated, and cereals especially wheat occupy a substantial share of this irrigated land (Belaqziz et al., 2014; Er-Raki et al., 2021). At the national scale, cereals extend over more than 5 million hectares (about 75% of arable land), with common wheat cultivated on approximately 2 million hectares and durum wheat on 0.9–1.0 million hectares. Under predominantly rainfed conditions, average wheat yields are around 2 t ha⁻¹ and rarely exceed 4 t ha⁻¹ even under irrigation, highlighting the combined impact of water deficit and sub-optimal management on productivity (Bouras et al., 2019; Belaqziz et al., 2021). Improving irrigation efficiency for wheat is therefore essential to stabilise yields and reduce pressure on limited water resources.

Despite advances in water management, traditional irrigation methods such as surface and flood irrigation remain dominant in Morocco (Bouchaou et al., 2011; Belaqziz et al., 2014; Alonso et al., 2019). These approaches are inefficient and poorly suited to semi-arid conditions with high evaporative demand, and often lead to substantial water losses through non-productive evaporation and deep percolation, as well as uneven spatial distribution of soil moisture (Schilling et al., 2012). Switching to modern systems, such as drip and automated irrigation, could reduce water use and improve yield stability (Er-Raki et al., 2021; Dutta et al., 2020). Recent technological innovations offer promising tools to enhance irrigation efficiency. Remote sensing, soil moisture sensors, and automated systems enable real-time water delivery tailored to crop needs, reducing waste and improving productivity (Ingrao et al., 2023; Lakhier et al., 2024; El Bouanani et al., 2025; Bourriz et al., 2025). Integrating these technologies into Moroccan agriculture could transform water-use practices and promote adaptation to more extreme climatic conditions.

In parallel, crop models have become essential for optimizing irrigation under data-scarce or water-limited conditions. Models such as Decision Support System for Agro technology Transfer (DSSAT) (Kephe et al., 2021) and Environmental Policy Integrated Climate (EPIC) (Lychuk et al., 2017), have been widely used to explore irrigation scenarios, evaluate yield responses, and support water management planning. Among them, AquaCrop, developed by FAO, is widely recognized for its simplicity and robustness in simulating crop responses to water stress across diverse environments (Vanuytrecht et al., 2014). AquaCrop model has been calibrated and validated for many crops, including maize, cotton, and quinoa, showing strong agreement between

simulated and observed yields and water use (Tan et al., 2018; Alvar-Beltrán et al., 2021; Feng et al., 2022; Barrera et al., 2024; Zhu et al., 2024). For wheat in Morocco and other semi-arid regions, previous studies have demonstrated the ability of AquaCrop to represent yield response to deficit irrigation and to assess alternative water-saving strategies (Iqbal et al., 2014; Toumi et al., 2016; Belaqziz et al., 2021; Zhang et al., 2022; Kaissi et al., 2024; Kaissi et al., 2025b).

In recent years, AquaCrop has increasingly been coupled with optimisation or data-assimilation methods to enhance its operational relevance. (Belaqziz et al., 2021) integrated remote sensing with AquaCrop to refine irrigation scheduling in a large surface-irrigated scheme, but the optimisation relied on relatively simple allocation rules and did not explicitly optimise stage-dependent thresholds under drip irrigation. (Zhang et al., 2022) applied the model to assess irrigation quotas under water-limited conditions. Similarly, (Guangshuai et al., 2022) evaluated soil moisture thresholds and irrigation methods for winter wheat using the AquaCrop model. In parallel, other studies have focused on exploring the use of standalone optimization algorithms, such as the Covariance Matrix Adaptation Evolution Strategy (CMA-ES), to improve irrigation allocation and scheduling in gravity irrigation systems, without fully exploiting the process-based structure of AquaCrop or targeting dynamic, phenology-specific thresholds (Belaqziz et al., 2014). Parallel developments have used AquaCrop in combination with particle swarm optimisation (PSO) for parameter calibration or data assimilation: for example, (Akbari et al., 2024) used PSO to calibrate soil parameters and simulate saffron growth based on high-resolution field measurements, while (Jin et al., 2020) assimilated canopy cover (CC) and biomass (B) derived from hyperspectral data into AquaCrop to improve maize yield estimation. (Sharafkhane et al., 2024) developed an AquaCrop plug-in-PSO framework to optimise maize irrigation schedules using in-season weather forecasts. These studies collectively demonstrate the potential of linking AquaCrop with optimisation or remote sensing, but most of them focus on parameter calibration, yield estimation, or forecast-based scheduling in maize, and few explicitly address multi-stage, threshold-based optimisation of drip irrigation in wheat under real farmer-managed conditions.

There is therefore still a need for studies that tightly couple AquaCrop-OSPy with advanced optimisation algorithms to derive and test dynamic irrigation thresholds explicitly tailored to crop phenology and drip systems, and evaluated directly against farmer practices in the field. This study aims to improve irrigation strategies for wheat production in semi-arid regions by integrating the AquaCrop-OSPy model with gradient-based optimisation techniques. Specifically, it assesses the added value of integrating the AquaCrop-OSPy model with two external gradient-based optimization techniques to refine irrigation scheduling through a constant-threshold NET strategy and a stage-specific soil moisture threshold (SMT) strategy. The overarching objective is to improve water productivity (WP) by reducing irrigation while maintaining crop yield. To this end, the study addresses three main questions: **(1)** How can gradient-based optimization techniques be integrated with AquaCrop-OSPy to improve the efficiency of irrigation scheduling? **(2)** How do real and optimized irrigation strategies differ in terms of WP, crop stress, and crop production? **(3)** What added value does coupling AquaCrop-OSPy with external optimization algorithms offer in guiding best irrigation practices for semi-arid wheat production systems? Addressing these questions enhances understanding of how optimized irrigation strategies can balance water savings and crop performance, leading to more informed decision-making and adaptive water management in semi-arid farming systems.

2. Materials and methods

2.1. Study area

This study was conducted on a private farm in Chichaoua, located in the Haouz Plain in the Marrakech-Safi region of Morocco. The Haouz

plain, one of the country's main agricultural zones, lies within the Tensift watershed at an elevation of 550 m, covering approximately 6000 km², of which 2000 km² are irrigated. The region is characterized by a semi-arid Mediterranean climate with highly variable temperatures throughout the year, ranging from hot summers, with average daily maximum temperatures of around 35°C in July and August, to cold winters with average minimum temperatures as low as 5°C in January (Abourida et al., 2008). Annual precipitation averages 250 mm, with approximately 85% of the total falling between October and April. Reference evapotranspiration (ET₀) is significantly higher, averaging approximately 1600 mm/year (Jarlan et al., 2015), indicating a substantial water deficit that necessitates the use of efficient irrigation practices for successful crop production.

This study focused on two winter wheat fields: Field 1 (F1: stressed conditions) and Field 2 (F2: normal conditions) monitored over two consecutive growing seasons, 2016–2017 (Season 1, S1) and 2017–2018 (Season 2, S2). Each field, covering approximately 1.5 ha, has a soil texture composed of 30% clay, 30% silt, and 40% sand (Rafi et al., 2024). Both fields were managed using drip irrigation and had the same sowing and harvesting dates. Sowing was carried out in late November, and harvesting was conducted in late May to early June (Ouaadi et al., 2023).

Despite similar soil and management conditions, the two fields exhibited contrasting irrigation regimes under farmer practices. F1 was classified as a “stressed” field because seasonal irrigation amounts were lower and several irrigation events were applied late relative to peak crop water demand, which led to visible stress symptoms (temporary wilting, reduced canopy cover) and reduced yield. F2 represented “normal” conditions, with higher and more frequent irrigation

applications that were better synchronized with crop water requirements and did not show pronounced water stress during the main growth stages.

To simplify reference throughout the study, the combinations of field and season are denoted as F1-S1 and F1-S2 for Field 1 during the 2016/2017 and 2017/2018 seasons, respectively, and F2-S1 and F2-S2 for Field 2 during the same periods. Fig. 1 provides an overview of the monitored fields and their location within the Haouz Plain.

2.2. Field instrumentation and measurements

Both fields were instrumented with a comprehensive monitoring setup to provide continuous hydrometeorological and crop data for AquaCrop calibration, validation, and WP analysis. ET_a was measured using an eddy-covariance station installed at 2.58 m and equipped with a CSAT3 sonic anemometer and a KH₂O krypton hygrometer, with 20 Hz raw data processed and corrected following Ezzahar et al. (2009); Rafi et al., (2019) and Bouswir et al., (2025). Soil moisture was monitored at multiple depths using TDR probes (CS615/CS655) calibrated against gravimetric soil samples. A nearby automatic weather station provided half-hourly measurements of radiation, air temperature, relative humidity, wind speed, and precipitation, which were used to estimate ET₀ with the FAO-56 Penman–Monteith method.

Crop development was monitored using hemispherical photography to quantify CC, following (Duchemin et al., 2006) and (Khabba et al., 2009). Aboveground biomass was measured by destructive sampling within representative quadrats at key phenological stages, and samples were oven-dried to constant weight to derive dry biomass per unit area. Grain yield was determined at harvest by cutting and threshing central

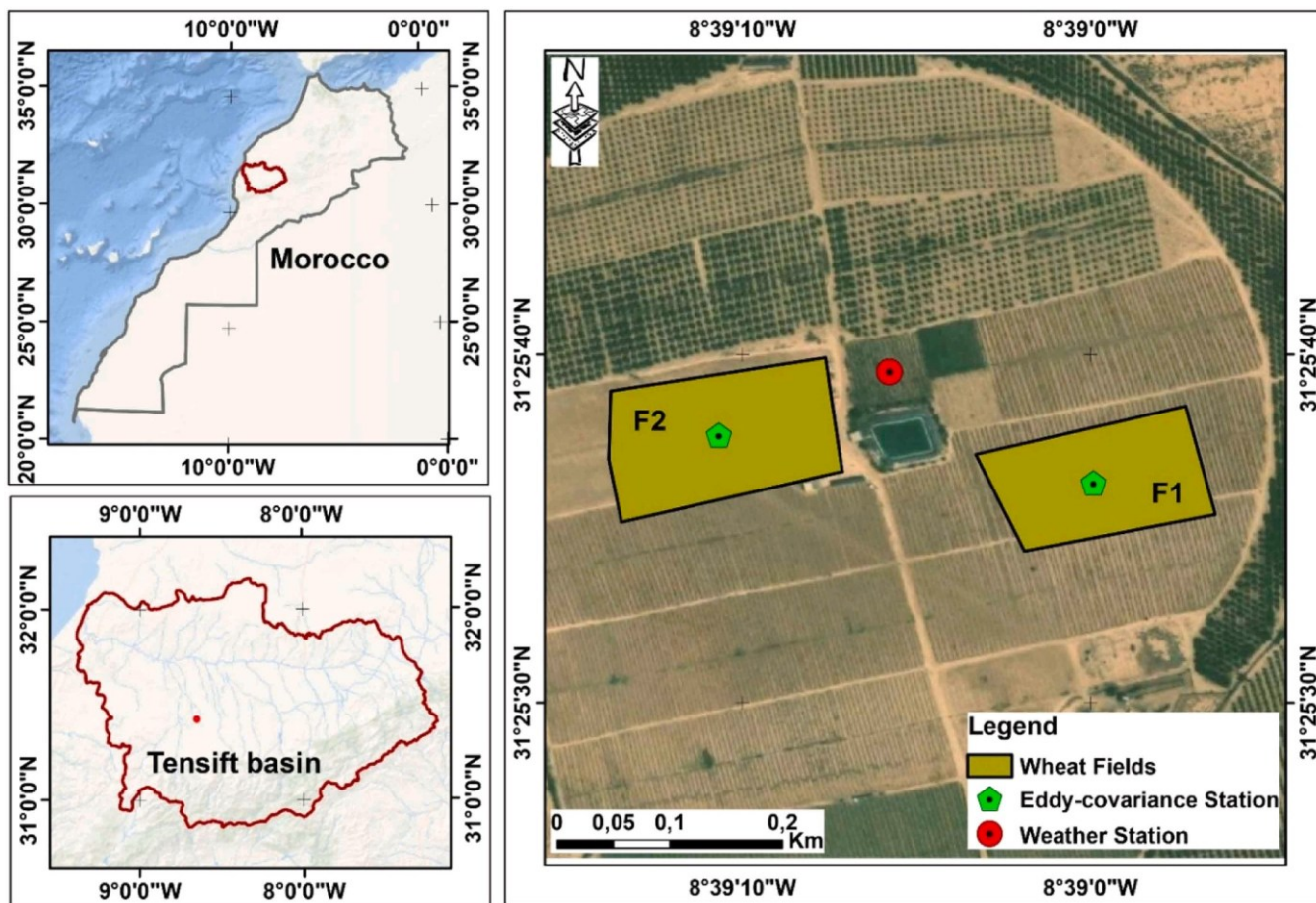


Fig. 1. Overview of the monitored fields in the Haouz Plain, Central Morocco.

rows in each field and converting the measured grain mass to yield ($t\ ha^{-1}$). Vegetation status was further characterized using a SKYE multi-spectral sensor (652 nm, 864 nm) to compute NDVI as a proxy for photosynthetically active biomass. These measurement procedures, fully described in previous studies on the same site (Rafi et al., 2019; Ouaadi et al., 2023), provided the CC, biomass, ETa, and yield datasets used in this study.

2.3. Data collection

AquaCrop-OSPy requires four main categories of input data to simulate crop growth and soil–water processes: weather conditions, soil properties, crop characteristics and management practices. In this study, all inputs were collected or derived specifically for the two wheat fields to ensure accurate representation of local conditions. In the following section, we will detail the essential input data required for optimal operation of the AquaCrop-OSPy model in our study area.

2.3.1. Weather data

AquaCrop requires daily meteorological inputs, including minimum and maximum temperatures (T_{min} , T_{max} [$^{\circ}C$]), precipitation (P [mm]), and reference evapotranspiration (ET_0 [mm/day]). The climatic conditions for both seasons are shown in Fig. 2A–B.

During the S1 season (Fig. 2A), temperatures ranged from $1.25^{\circ}C$ to $43.9^{\circ}C$, with limited and irregular rainfall reaching a daily maximum of 14.07 mm. Evaporative demand was high, with ET_0 peaking at 10.35 mm/day, creating strong water-stress potential and making crops highly dependent on irrigation (Belagziz et al., 2021).

During S2 (Fig. 2B), temperatures were slightly moderated ($-1.19^{\circ}C$ to $40.67^{\circ}C$), and rainfall events were more frequent, with daily amounts up to 30.15 mm. Despite this increase, overall moisture availability remained constrained due to persistent drought conditions, and ET_0

ranged from 0.58 to 8.99 mm/day. These climatic patterns highlight the strong interannual variability in crop water demand and reinforce the need for adaptive irrigation scheduling in the study area.

2.3.2. Soil data

In our study area, the soil is characterized by a texture of around 30% clay, 30% silt, and 40% sand, typical of the agricultural soils of the Haouz plain (Rafi et al., 2024). This texture impacts water retention and infiltration rates, with clay having a wider capacity to retain moisture, while sand facilitates drainage. Soil samples were collected from both fields to determine the key hydraulic properties required by AquaCrop, namely soil water content at field capacity (θ_{FC}), permanent wilting point (θ_{WP}), saturation (θ_{SAT}), bulk density (BD), and saturated hydraulic conductivity (K_{sat}). Hydraulic properties were estimated from measured soil texture and bulk density using the pedotransfer equations of (Wösten et al., 1999). These properties were then used to compute total available water ($TAW = \theta_{FC} - \theta_{WP}$) and to parameterize the layered soil profile in AquaCrop-OSPy. The resulting set of textural and hydraulic parameters, summarized in Table 1, provided the basis for representing soil water storage and flow in the model and ensured a realistic simulation of soil moisture dynamics under the different irrigation strategies tested in this study.

Within each field, soil spatial variability was handled by combining a detailed vertical characterization of the soil profile with a representative description at the plot scale. Calibrated TDR sensors installed at several depths (typically between 5 and 50 cm) provided continuous measurements of volumetric soil water content. These measurements were aggregated into root-zone soil water content using a depth-weighted averaging approach and then converted into available soil water following standard procedures described for the same experimental site (Rafi et al., 2019; Ouaadi et al., 2023). Along with soil texture and hydraulic properties (θ_{FC} , θ_{WP} , $Z_{r,max}$, TAW) summarized in Table 1, these

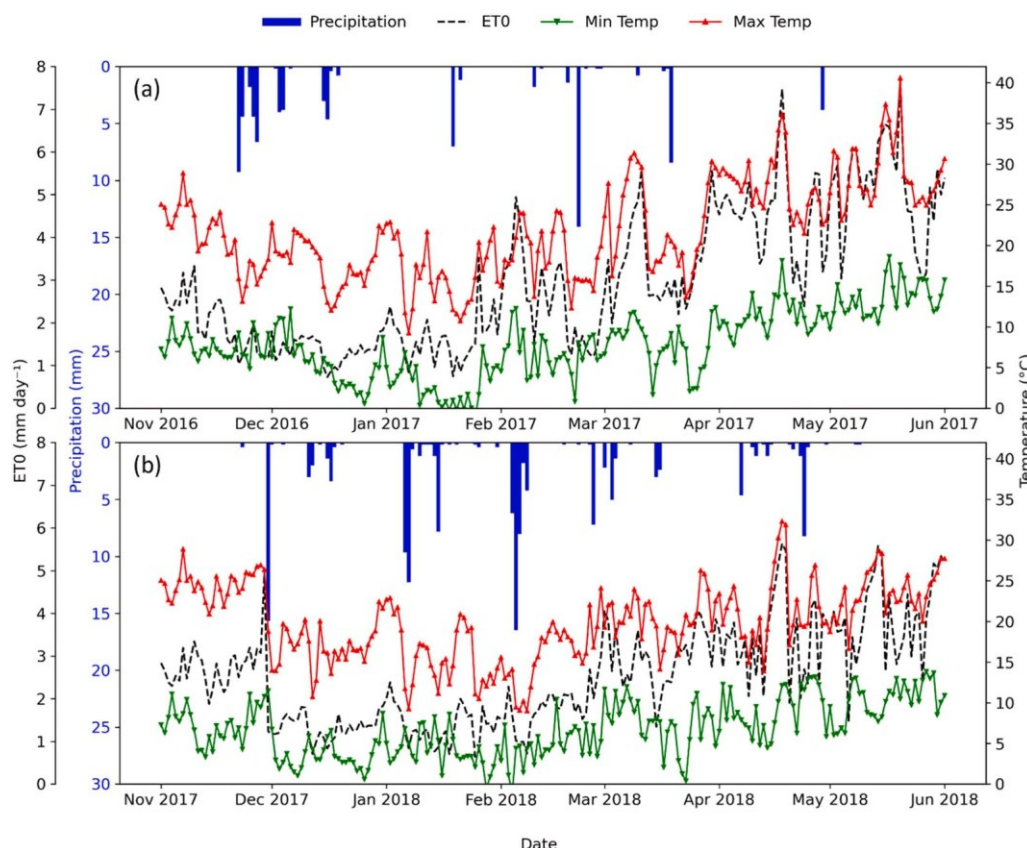


Fig. 2. Daily evolution of meteorological parameters during 2016/2017 (a) and 2017/2018 (b) growing seasons.

Table 1
Soil textural and hydraulic properties of the study fields.

Property	Value	Unit	Description
Soil texture	Clay Loam	-	-
Sand content	30	%	-
Silt content	30	%	-
Clay content	40	%	-
Permanent wilting point	0.17	cm ³ cm ⁻³	Soil water content at which plants cannot extract water
Field capacity (θ_{fc})	0.29	cm ³ cm ⁻³	Soil water content after free drainage
Saturation (θ_{sat})	0.49	cm ³ cm ⁻³	Maximum soil water content
Bulk density (BD)	1.34	g cm ⁻³	Soil compaction level
Saturated hydraulic conductivity (Ksat)	3.41	mm h ⁻¹	Water transmission rate in saturated conditions
Available water capacity (TAW)	0.127	cm ³ cm ⁻³	Amount of extractable water (FC–WP)

data were used to define a representative layered soil profile for F1 and F2. Given the small size, flat topography, and uniform drip-irrigation management of the plots, this profile was assumed to be laterally homogeneous within each field, so that vertical variability is explicitly represented while lateral variability is aggregated at the field scale.

2.3.3. Irrigation data

Irrigation input data, such as application timing, frequency, and volumes, were collected across both seasons to aid in the calibration and validation of AquaCrop. All irrigation was conducted via drip systems. To evaluate the performance of various irrigation strategies, three scenarios were simulated in AquaCrop, ranging from traditional practices to optimized approaches. This setup enabled a comparative analysis of WP, yield, and resource efficiency. Both fields were irrigated using drip systems, with individual applications of approximately 12.5 mm per event (Ouaadi et al., 2021). Field 2 served as the fully irrigated reference, where irrigation was scheduled to meet crop water requirements throughout the season. Field 1, in contrast, was subjected to drought stress during the reproductive stage, with irrigation withheld between February 27 and March 14, 2017, to induce a controlled water deficit. This treatment resulted in lower evapotranspiration compared to Field 2, while all other management practices, including sowing date, wheat variety, fertilization, and plant density, were kept identical.

The first strategy is Real Field Irrigation, which represents local farmer-managed irrigation practices. Drip irrigation was used, and water was applied based on the farmer's judgment of crop water requirements, relying on visual indicators such as leaf wilting and soil appearance. No decision support tools, soil moisture sensors, or modeling techniques were used. Irrigation events and quantities were recorded as they occurred during the S1 and S2 growing seasons, providing a baseline for comparison with optimized strategies.

In AquaCrop, irrigation scheduling for model-based strategies is governed by the soil water depletion fraction (p), which defines the fraction of Total Available Water (TAW) that can be depleted before water stress begins to affect crop growth. Root-zone water depletion (Dr) is evaluated relative to TAW, such that $Dr/TAW = 0$ at field capacity ($K_s = 1$) and $Dr/TAW = 1$ at the permanent wilting point ($K_s = 0$). Water stress starts when Dr/TAW exceeds p and intensifies as the soil continues to dry towards the lower limit of TAW. This threshold-based formulation (Vanuytrecht et al., 2014) is used in this study to trigger irrigation events in the NET and SMT strategies.

The following strategy, NET Irrigation, is a threshold-based approach that maintains soil water content at a constant level throughout the crop cycle. In AquaCrop-OSPy, NET operates by continuously monitoring root-zone depletion (Dr) and triggering irrigation whenever Dr exceeds a fixed fraction of the Total Available Water (TAW). When the depletion threshold is exceeded, AquaCrop automatically triggers an irrigation event that replenishes all soil compartments to the target moisture level,

effectively refilling the soil profile toward field capacity and restoring the crop to non-stress conditions ($K_s \approx 1$). NET irrigation does not vary with phenology: the same threshold is applied across all growth stages, and irrigation timing is controlled solely by the soil–water balance rather than the crop's developmental stage. As such, NET provides a simple yet robust rule-based strategy to maintain a stable soil-moisture regime while minimizing unnecessary deep percolation.

On the other hand, SMT Irrigation adopts a stage-specific strategy in which irrigation is triggered whenever the root-zone soil water content falls below a predefined depletion threshold, with four distinct thresholds corresponding to the major phenological stages of wheat (emergence, canopy expansion, maximum canopy, and senescence). Unlike NET, which applies a single constant threshold throughout the season, SMT adjusts its thresholds to match the crop's changing water requirements. This dynamic formulation enables irrigation to be aligned more closely with crop physiology: higher thresholds can be applied during sensitive phases such as canopy development and grain filling, whereas lower thresholds can be used during early or late growth. As a result, SMT provides a more responsive and targeted irrigation strategy aimed at improving yield stability and water use efficiency under semi-arid conditions. For both NET and SMT, a baseline depletion threshold of 60% of TAW was used, in accordance with the recommendations of (Toumi et al., 2016), ensuring a consistent and physiologically relevant reference point for wheat grown under semi-arid Moroccan conditions.

2.4. AquaCrop model

AquaCrop is a crop simulation model developed by the Food and Agriculture Organization (FAO) of the United Nations to evaluate the effects of soil water availability, irrigation scheduling, and management practices on crop productivity. It is particularly suitable for simulating the yield of herbaceous crops under water-limited conditions. Compared to other crop models, AquaCrop requires fewer parameters and simplified inputs, enhancing its usability and robustness in diverse agricultural environments.

Building on the success of the original model, FAO released AquaCrop-OSPy (ACOSP), an open-source Python implementation that enhances the model's flexibility and computational efficiency (Kelly and Foster, 2021). ACOSP is synchronized with AquaCrop version 6.1 and introduces several features that address user needs, particularly for advanced crop-water management and policy analysis. Designed for users with basic Python skills, ACOSP leverages Python's popularity and rich ecosystem, making it adaptable for tasks like parameter calibration, sensitivity analysis, and scenario modeling.

AquaCrop simulates soil water balance, crop development, and yield on a daily time step using crop, soil, weather, and management input data. Its structure separates E_{ta} into crop transpiration (Tr) and soil evaporation (E), improving the simulation of water use and biomass production. Tr is estimated from ET_0 , scaled by CC and adjusted by a stress coefficient (K_s), while E is driven by surface soil moisture and the fraction of uncovered soil. ET_0 is computed using the FAO Penman-Monteith method. Moreover, the model normalizes WP relative to ET_0 and CO_2 concentration, enabling consistent performance across different climates. WP reflects the efficiency of crop water use, expressed as the amount of yield produced per unit of water consumed. It also accounts for dynamic environmental stressors such as water deficits, temperature variation, and elevated CO_2 levels, making it a powerful tool for assessing climate change impacts on crop yield (Bouras et al., 2019).

2.5. Methodology

This study employs a structured and integrated approach to optimize irrigation practices and improve crop yields by combining a process-based model with an optimization algorithm. The methodology is structured into several key phases, each building on the previous one to ensure a strong analysis (Fig. 3). The process begins with acquiring and

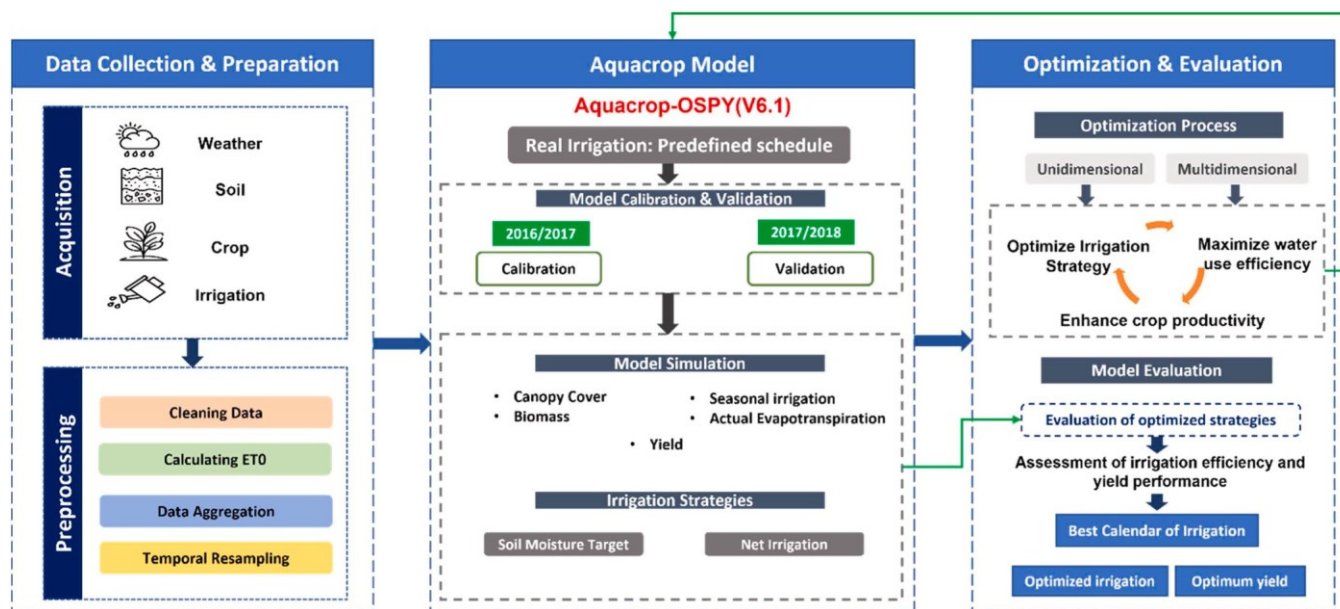


Fig. 3. Flowchart describing the general methodology employed in this study.

preparing essential climatic and agronomic data, including temperature, precipitation, soil characteristics, and crop-specific parameters, which are crucial for accurately simulating crop growth and water use under local conditions. After data collection, pre-processing steps such as handling missing values, correcting outliers, and resampling are performed to ensure data quality for modeling. Building on this foundation, the AquaCrop-OSPy model (version 6.1) was calibrated using field data collected during the 2016/2017 growing season and validated during the 2017/2018 growing season. With the model calibrated and validated, the simulations of three distinct irrigation scenarios: Real Field Irrigation, NET Irrigation, and SMT Irrigation, were conducted to assess the relative impacts of traditional and optimized irrigation practices on water use efficiency and crop yield. Following the simulation phase, an optimization process is implemented to identify the most efficient irrigation schedule. This optimization aims to maximize yield while minimizing water use, focusing on factors such as WP and overall crop performance. The process involves systematically adjusting irrigation parameters to select the best strategy based on criteria to achieve maximum yield and efficient water use. Finally, the optimized irrigation strategies are evaluated to confirm their effectiveness in balancing irrigation needs and yield optimization. This evaluation provided practical, data-driven insights and recommendations for improving irrigation practices and water management in agriculture.

2.6. Calibration and validation of AquaCrop model

The AquaCrop-OSPy (ACOSP) model was calibrated and validated using field data collected over two successive growing seasons to ensure reliable simulations of wheat development and yield under local semi-arid conditions. Calibration was carried out using data from the 2016/2017 season, during which model parameters were adjusted to align simulated outputs with observed values for CC, B, ETa, and yield (Y).

To achieve this, the calibration process focused on non-conservative parameters, such as maximum canopy cover (CCx), canopy growth and decline coefficients (CGC and CDC), crop transpiration coefficient (KcTr, x), reference harvest index (HI0), and normalized water productivity (WP*). In contrast, conservative parameters, including base temperature (Tbase) and upper temperature (Tupper), were kept fixed, as they are assumed to remain stable across different environments and management practices.

The selection of parameter ranges was guided by the AquaCrop user

manual (Raes et al., 2009) and was further informed by values reported in the literature, particularly in the study by (Toumi et al., 2016), who calibrated AquaCrop for wheat under similar Moroccan semi-arid conditions. However, unlike (Toumi et al., 2016), who used a manual calibration approach, this study implemented a grid search method (Verwaeren et al., 2015), which systematically explored combinations of key parameters to identify the best alignment between simulated and observed crop behavior.

The validation step was performed using data from the S2 season, applying the same calibrated parameter set without modification. This process allowed for testing the model's transferability to different seasonal conditions within the same region. The final parameter values used in both calibration and validation are summarized in Table 2, and they represent a robust configuration for simulating wheat performance under drip irrigation in the Haouz Plain. The calibration and validation steps confirmed AquaCrop-OSPy's ability to simulate wheat growth reliably under local semi-arid conditions, providing a solid basis for the evaluation of alternative irrigation strategies.

2.7. Water productivity

WP measures water use efficiency in crop production and is typically defined as the ratio of crop yield to the total water consumed. It is a crucial metric in evaluating how effective water resources are utilized in agriculture. WP is particularly important in regions facing water scarcity, as it helps optimize water use to increase agricultural production while minimizing wastage. In this study, WP was evaluated using two different methods, both of which are based on the AquaCrop model to assess water efficiency under various irrigation scenarios. These methods provide a better understanding of how water is used in crop production.

The first method calculates water consumption by taking the ratio between Y and ETa, which reflects the efficiency of water use while considering losses due to transpiration and evaporation. This method is described by the following equation:

$$WP_{ETa} = \frac{Y}{ETa} \quad (1)$$

Where Y is Crop yield (kg or tons per hectare) and ETa is Actual evapotranspiration (mm).

The second method focuses on irrigation water productivity

Table 2

Key conservative and non-conservative parameters used for AquaCrop model in wheat production.

Parameters			Value	Unit	Calibration Status
	Variables	Description			
Conservative	Crop	Crop Type	Wheat	-	Default
	Tbase	Base temperature	5	°C	Literature
	Tupp	Upper temperature	33	°C	Literature
	WP*	Normalized crop Water Productivity	16	g/ m ²	Calibrated
	Kc Tr,x	Maximum coefficient for transpiration	1.07	-	Calibrated
	K_{ex}	Maximum coefficient for soil evaporation	0.25	-	Calibrated
	HIO	Reference harvest Index	46	%	Calibrated
	P_{exp,upper}	Upper threshold for canopy expansion	0.3	-	Default
	P_{exp,lower}	Lower threshold for canopy expansion	0.8	-	Default
	CCO	Initial canopy cover	5.06	%	Calibrated
	Zmin	Minimum Effective Rooting Depth	0.1	m	Default
	Zmax	Maximum Effective Rooting Depth	0.55	m	Literature
	EMERGENCE	Growing degree/Calendar days from sowing to emergence/transplant recovery	82	CGDD	Calibrated
	MATURITY	Growing degree/Calendar days from sowing to maturity	1462	CGDD	Calibrated
Non-Conservative	MAXROOTING	Growing degree/Calendar days from sowing to maximum rooting	696	CGDD	Calibrated
	SENESCENCE	Growing degree/Calendar days from sowing to senescence	972	CGDD	Calibrated
	CCx	Maximum canopy cover (fraction of soil cover)	0.95	%	Calibrated
	CDC	Canopy decline coefficient (fraction per GDD/calendar day)	0.60	%/CGDD	Calibrated
	CGC	Canopy growth coefficient (fraction per GDD)	0.89	%/CGDD	Calibrated

($WP_{Irrigation}$), which evaluates the efficiency of irrigation water use. It calculates crop yield with the amount of irrigation water applied to the exclusion of other water sources such as precipitation. The $WP_{Irrigation}$ formula is as follows:

$$WP_{Irrigation} = \frac{Y}{I} \quad (2)$$

Where Y is Crop yield (kg or tons per hectare) and I is irrigation water applied (mm).

These parameters were obtained from AquaCrop simulations, providing a comprehensive assessment of water use efficiency. This analysis allows the identification of optimal irrigation strategies that improve crop yields while conserving water resources.

2.8. Optimization method

To improve irrigation scheduling, this study implements two specific optimization methods within the AquaCrop-OSPy model: a unidimensional optimization using the minimize_scalar function with the bounded method and multidimensional optimization using the minimize function with the L-BFGS-B algorithm (Zhu et al., 1997). In both cases, the decision variables are the irrigation depletion thresholds, and the goal is to identify threshold values that enhance $WP_{Irrigation}$ while preserving satisfactory yield levels.

To clarify the optimization framework, the objective function is explicitly defined in terms of $WP_{Irrigation}$. For each irrigation strategy, the optimizer searches for the threshold vector θ that maximizes WP, expressed as the ratio between simulated yield and total seasonal irrigation:

$$\max_{\theta} J(\theta) = WP(\theta) = \frac{Y(\theta)}{I(\theta)} \quad (3)$$

Where $Y(\theta)$ is the simulated yield (t ha⁻¹), and $I(\theta)$ is the total seasonal irrigation (mm) produced by AquaCrop-OSPy under a given set of thresholds θ . This formulation directly captures the trade-off between water use and yield, favouring threshold combinations that produce more grain per unit of irrigation water.

Since minimize_scalar and minimize are designed to solve minimization problems, the optimization is implemented numerically by minimizing the negative of the objective function:

$$\min_{\theta} -J(\theta) = -WP(\theta) \quad (4)$$

For the NET strategy, the decision vector reduces to a single

depletion threshold, $\theta = [\theta_{Net}]$. For the SMT strategy, the decision vector is defined as $\theta = [\theta_1, \theta_2, \theta_3, \theta_4]$, corresponding to the four phenological stages. This formulation provides a transparent theoretical basis for the optimization process.

In both optimization schemes, only simple bound constraints were imposed on the decision variables. The NET threshold and the SMT thresholds were restricted to physically realistic ranges (0–100% of TAW), but no additional linear or non-linear constraints were specified in the optimizer. Although a maximum seasonal irrigation (Max-IrrSeason) was considered during the design phase to define plausible parameter ranges, it was not implemented as an explicit hard constraint in the final optimization runs. Instead, seasonal irrigation volumes resulted directly from the AquaCrop simulations and were implicitly accounted for through the WP-based objective function

2.8.1. Unidimensional optimization (NET irrigation threshold - NET)

This method optimizes a single decision variable, the NET irrigation threshold (NetIrrSMT), using the minimize_scalar function with the bounded method. This approach enables the precise alignment of irrigation events with crop requirements, improving efficiency compared to fixed-schedule or experience-based irrigation methods. The bounded method was selected for its ability to restrict the parameter search within realistic limits, ensuring feasible outcomes and faster convergence. Alternative algorithms, such as Nelder-Mead and Trust Region Reflective, were tested but showed lower computational efficiency and slower convergence, making them less suitable for this single-parameter optimization (Fig. 4).

2.8.2. Multidimensional optimization (Soil moisture target - SMT)

This approach refines multiple soil moisture thresholds across crop phenological stages, enabling a dynamic and responsive irrigation schedule. Optimization was performed using the limited-memory Broyden-Fletcher-Goldfarb-Shanno (L-BFGS-B) algorithm, a gradient-based method well-suited for problems involving multiple interdependent variables (Fig. 5). L-BFGS-B was chosen for its ability to efficiently capture complex relationships between soil moisture levels and crop development, offering faster convergence compared to heuristic methods like Genetic Algorithms (GA) or Particle Swarm Optimization (PSO), which are more computationally demanding and less practical for continuous-variable problems.

These two approaches illustrate the complementary nature of single-variable and multi-variable optimization in irrigation management. The unidimensional method ensures localized adjustments to specific parameters, while the multidimensional method accounts for the intricate

ALGORITHM 1: OPTIMIZATION OF NET IRRIGATION THRESHOLD (NET)**Input:** Initial NET Irrigation Threshold Maximum Seasonal Irrigation**Output:** Optimized NET Irrigation Threshold (NetIrrSMT_opt)

```

1  Define the multi-objective function:
2      Maximize Yield (Y) while minimizing Total Irrigation (Irr).
3      Compute Water Productivity (WP) = Y / Irr.
4      Objective: Maximize WP.
5  Initialize optimization:
6      Set bounds for NetIrrSMT (e.g., 0% to 100% of TAW).
7      Select optimization method: minimize_scalar with the "bounded" method.
8  Optimization process :
9      For each NetIrrSMT within the bounds:
10         Run the AquaCrop model with NetIrrSMT.
11         Retrieve Yield (Y) and Total Irrigation (Irr).
12         Compute WP = Y / Irr.
13         Store the result.
14  Select the optimal NetIrrSMT :
15      Identify the threshold that maximizes WP.
16  Output the optimized NetIrrSMT_opt.
17  End

```

Fig. 4. Pseudocode representation for Optimizing the Net Irrigation Threshold in AquaCrop-based irrigation scheduling.

interplay of multiple factors influencing water use efficiency. Building on the work of (Toumi et al., 2016), which proposed a depletion threshold (Dr) of 0.6 for irrigation management, this study refines this threshold to better align with the specific environmental and management conditions of the study area. The depletion threshold (Dr) refers to the proportion of the soil's available water that can be depleted before water stress starts to negatively affect crop growth.

In practice, AquaCrop-OSPy was treated as a black-box simulator mapping a set of four SMT thresholds to a scalar objective function WP. The L-BFGS-B algorithm was implemented via the minimize function of the SciPy library, which computes gradients numerically using finite-difference approximations of WP with respect to the SMT variables based on repeated model evaluations. We used the default convergence tolerances and iteration limits of the SciPy L-BFGS-B implementation, with bound constraints of 0–100% of TAW on each SMT threshold to keep the search within agronomically realistic ranges.

2.9. Model evaluation

To evaluate AquaCrop performance under semi-arid field conditions, both qualitative and quantitative analyses were conducted using data from F1-S1, F1-S2, F2-S1, and F2-S2. The qualitative evaluation involved graphical comparisons between simulated and observed values, allowing visual assessment of temporal patterns, irrigation scheduling, and general agreement across different crop growth stages.

For quantitative evaluation, the accuracy of the model was assessed using several statistical metrics. The mean square error (MSE) (Eq. (5)) and root mean squared error (RMSE) (Eq. (6)) were calculated to quantify the average error in model prediction, providing a measure of the variation of simulated data from observed values. In addition, the coefficient of determination (R^2) (Eq. (7)) was used to determine the proportion of variance in the dependent variable that is explained by the independent variables, indicating the model's reliability. Two additional performance metrics were also considered: the Index of Agreement (d) (Eq. (8)), which assesses the degree to which model predictions approach observed values, scaled to the variability of the observed data, and the Nash–Sutcliffe Efficiency (NSE) (Eq. (9)), which evaluates how well model predictions reproduce the observed time series relative to the mean of observations. These metrics provided a strong assessment of the model's performance, highlighting its effectiveness and points for improvement in simulating agricultural results under various conditions.

$$MSE = \frac{1}{n} \sum_{i=1}^n (O_i - S_i)^2 \quad (5)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (O_i - S_i)^2} \quad (6)$$

ALGORITHM 2: OPTIMIZATION OF SOIL MOISTURE TARGETS (SMT)

Input: Initial SMT thresholds [SMT_1, SMT_2, SMT_3, SMT_4], Maximum Seasonal Irrigation (MaxIrrSeason)

Output: Optimized SMT thresholds (SMT_opt)

- 1 Define the multi-objective function:
 - 2 Maximize Yield (Y) while minimizing Total Irrigation (Irr).
 - 3 Compute Water Productivity (WP) = Y / Irr.
 - 4 Objective: Maximize WP.
 - 5 Initialize optimization:
 - 6 Set bounds for **SMT at each growth stage**:
 - 10 •SMT_1 (Emergence)
 - 11 •SMT_2 (Canopy Growth)
 - 12 •SMT_3 (Grain Filling)
 - 13 •SMT_4 (Senescence)
 - 14 Use **minimize** with **L-BFGS-B algorithm** for constrained optimization.
 - 15 Optimization process :
 - 16 For each combination of SMT thresholds:
 - 17 Run the **AquaCrop model** with SMT values.
 - 18 Retrieve **Y and Irr**.
 - 19 Compute WP = Y / Irr.
 - 20 Store the result.
 - 21 Select the optimal SMT values:
 - 22 Identify the **SMT thresholds that maximize WP**.
 - 23 Output the optimized **SMT values (SMT_opt)**.
 - 24 **End**
-

Fig. 5. Pseudocode representation for Optimizing Soil Moisture Target Thresholds based on crop growth stages.

$$R^2 = \frac{[\sum_{i=1}^n (O_i - \bar{O})(S_i - \bar{S})]^2}{[\sum_{i=1}^n (O_i - \bar{O})^2][\sum_{i=1}^n (S_i - \bar{S})^2]} \quad (7)$$

$$d = 1 - \frac{\sum_{i=1}^n (O_i - S_i)^2}{\sum_{i=1}^n (|S_i - O| + |O_i - O|)^2} \quad (8)$$

$$NSE = 1 - \frac{\sum_{i=1}^n (O_i - S_i)^2}{\sum_{i=1}^n (O_i - O)^2} \quad (9)$$

Where O_i is the observed value, S_i is the simulated value, $\bar{O}$ and $\bar{S}$ are the means of observed and simulated values, respectively, and n is the number of observations. NSE values range from $-\infty$ to 1, with NSE > 0.5 generally indicating acceptable model performance, while the Index of Agreement d ranges from 0 to 1, with values close to 1 indicating strong agreement between simulated and observed data.

2.10. Sensitivity analysis

To assess the robustness and potential generalizability of the calibrated AquaCrop-OSPy parameters, a one-at-a-time (OAT) sensitivity analysis was performed on the variables most likely to influence model outputs. Seven parameters were evaluated: five crop parameters (normalized water productivity for biomass, WP^* , harvest index at flowering, HI_o , maximum rooting depth, Zr_{max} , and the canopy growth coefficient, CGC) and two soil hydraulic properties (the soil water contents at field capacity, θ_{FC} , and permanent wilting point, θ_{WP}). Each parameter was perturbed by $\pm 20\%$ around its calibrated value, following established AquaCrop sensitivity protocols (Vanuytrecht et al., 2014; Lu et al., 2021; Jallal et al., 2025).

The analysis was carried out for two representative situations (F1–S1 and F2–S2) and for three contrasting irrigation regimes: farmers' Real practice, optimized NET (NETopt), and optimized SMT (SMTopt). For each perturbed value, AquaCrop-OSPy was run and the resulting grain yield (Y) was recorded. Parameter sensitivity was quantified as:

$$\text{Sensitivity}(\%) = \frac{1}{2} \left[\frac{|Y_{\text{low}} - Y_{\text{base}}|}{Y_{\text{base}}} + \frac{|Y_{\text{high}} - Y_{\text{base}}|}{Y_{\text{base}}} \right] * 100 \quad (10)$$

where Y_{base} is the model output with calibrated parameters and Y_{low} , Y_{high} to the yields obtained with -20% and $+20\%$ perturbations, respectively.

This procedure makes it possible to (i) identify the parameters exerting the strongest influence on simulated yield and (ii) examine whether the relative ranking of influential parameters remains stable across Real, NETopt, and SMTopt irrigation. A stable ranking across irrigation regimes suggests that the calibrated parameters represent general physiological behaviour rather than scenario-specific artefacts, thereby increasing confidence in the reliability of the model for scenario analysis.

3. Results

3.1. Calibration and validation of the AquaCrop model

As summarized in Table 3, the AquaCrop model exhibited strong performance in simulating CC, B, and ETa under local semi-arid conditions.

During calibration (S1, 2016/2017), AquaCrop reliably captured the CC dynamics in both fields, with R^2 values of 0.91 for F1 and 0.92 for F2, reflecting excellent agreement with observed trends across growth stages. This is further supported by NSE values of 0.91 (F1) and 0.95 (F2) and indices of agreement d of 0.98 and 0.96, respectively. Biomass simulation also showed strong agreement, with R^2 values of 0.86 (F1) and 0.82 (F2), highlighting the model's capacity to accurately predict biomass accumulation under actual irrigation conditions, the corresponding NSE values (0.86 and 0.82) and d values (0.98 and 0.94) confirm this good performance. The ETa simulation, although presenting slightly higher variability, maintained strong agreement with observed values ($R^2 = 0.75$ for F1 and 0.84 for F2), with NSE values of 0.73 and 0.77 and d values of 0.91 and 0.93, respectively. The model effectively estimated evapotranspiration patterns, despite potential fluctuations resulting from variations in atmospheric demand and soil moisture availability.

For the validation phase (S2, 2017/2018), AquaCrop continued to demonstrate robust performance. CC simulation was highly consistent with observed values, with R^2 values of 0.93 (F1) and 0.91 (F2), reflecting its strong ability to replicate canopy development with minimal error, as supported by a low MSE of 0.01 and RMSE of 0.09–0.10, together with NSE values of 0.90 and 0.94 and d values of 0.97 and 0.98, respectively. Biomass predictions also remained accurate, with R^2 values

of 0.76 (F1) and 0.78 (F2), and only minor deviations at specific growth phases, this is consistent with NSE values of 0.78 (F1) and 0.76 (F2) and d values of 0.92 and 0.94. Although ETa showed slightly more variability, the R^2 values (0.74 for both fields) and RMSEs of 0.55 mm/day (F1) and 0.52 mm/day (F2), with NSE values of 0.73 (F1) and 0.71 (F2) and d values of 0.91 and 0.86, confirm the model's ability to capture seasonal evapotranspiration patterns. These deviations may be attributed to environmental factors such as fluctuating atmospheric demand and localized variations in soil moisture availability.

Although minor discrepancies were noted in ETa, attributed to environmental variability, the model's global performance remains strong, supporting its application in optimizing irrigation strategies. These results validate the applicability of AquaCrop for simulating wheat growth and water use dynamics, and they establish a solid foundation for subsequent evaluation of alternative irrigation strategies under water-limited conditions.

3.2. Challenges in farmer irrigation practices: Ks and water dynamics analysis

Although the AquaCrop model performed well during calibration and validation, its application under real field conditions revealed inefficiencies in irrigation management. To better understand the causes of suboptimal water, use and yield variability, we analyzed the Ks with irrigation and rainfall dynamics in both fields and growing seasons.

Farmer irrigation practices often lack precision, resulting either in stress during critical stages or in over-irrigation and deep percolation losses. As shown in Fig. 6, the Ks dynamics, combined with irrigation and precipitation data, highlight these challenges for both fields during the S1 and S2 growing seasons. In AquaCrop, the water stress coefficient (Ks) quantifies the reduction in crop transpiration under water-limiting conditions, ranging from 1 (no stress) to 0 (complete stress).

For example, in F1-S1, Ks frequently drops below 1 during key growth stages, such as canopy development and grain filling, indicating insufficient irrigation that does not meet the crop's water requirements. This prolonged water stress negatively impacts both biomass accumulation and yield. Similarly, F1-S2 shows comparable periods of stress, further highlighting the inappropriate use of irrigation during critical stages of wheat growth.

In addition to water stress, over-irrigation is another challenge, illustrated by the events of excessive irrigation visible in F2-S1 and F2-S2. The green bars representing irrigation occasionally exceed the crop's water requirements, resulting in water loss by deep percolation beyond the root zone, making it unavailable to the plant. These periods of over-irrigation lead to inefficient water use, lower WP and higher irrigation costs. In addition, there's a time lag between irrigation and precipitation, particularly in zones F1-S2, where irrigation episodes overlap with or immediately follow precipitation. This redundancy highlights a lack of synchronization between irrigation schedules and rainfall patterns, leading to further inefficiencies in water use.

Field-specific challenges are also evident in the analysis. For example, F1 experienced more frequent and longer declines in Ks than F2, suggesting that field F1 is more sensitive to water stress due to irrigation scheduling or soil characteristics. This disparity highlights the need for adapted irrigation practices that consider the specific conditions of each field. Collectively, these challenges underline the importance of optimizing irrigation thresholds and strategies to address water stress, avoid over-irrigation and better align irrigation with precipitation. These improvements are essential to increase water use efficiency and maintain consistent yields. This analysis provides the basis for the implementation of optimized irrigation practices, which will be examined in greater detail in the next section.

These observations show that even with a well-calibrated and validated model like AquaCrop, real-world irrigation practices often do not match crop water requirements. Frequent episodes of stress, over-irrigation and poor synchronization with rainfall reveal a mismatch

Table 3

Statistical Evaluation of Observed and Simulated Crop Parameters during Calibration (2016/2017) and Validation (2017/2018) under Real Irrigation.

Crop Parameters	Metrics	Season 1: 2016/2017		Season 2: 2017/2018	
		F1	F2	F1	F2
Canopy Cover (-)	R^2	0.91	0.92	0.93	0.91
	MSE	0.01	0.01	0.01	0.01
	NSE	0.91	0.95	0.90	0.94
	d	0.98	0.96	0.97	0.98
	RMSE	0.10	0.10	0.09	0.10
Biomass (tonne/ha)	R^2	0.86	0.82	0.76	0.78
	MSE	0.02	0.03	0.19	0.12
	NSE	0.86	0.82	0.78	0.76
	d	0.98	0.94	0.92	0.94
	RMSE	0.15	0.18	0.44	0.35
Actual Evapotranspiration (mm)	R^2	0.75	0.84	0.74	0.74
	MSE	0.37	0.31	0.31	0.27
	NSE	0.73	0.77	0.73	0.71
	d	0.91	0.93	0.91	0.86
	RMSE	0.61	0.56	0.55	0.52

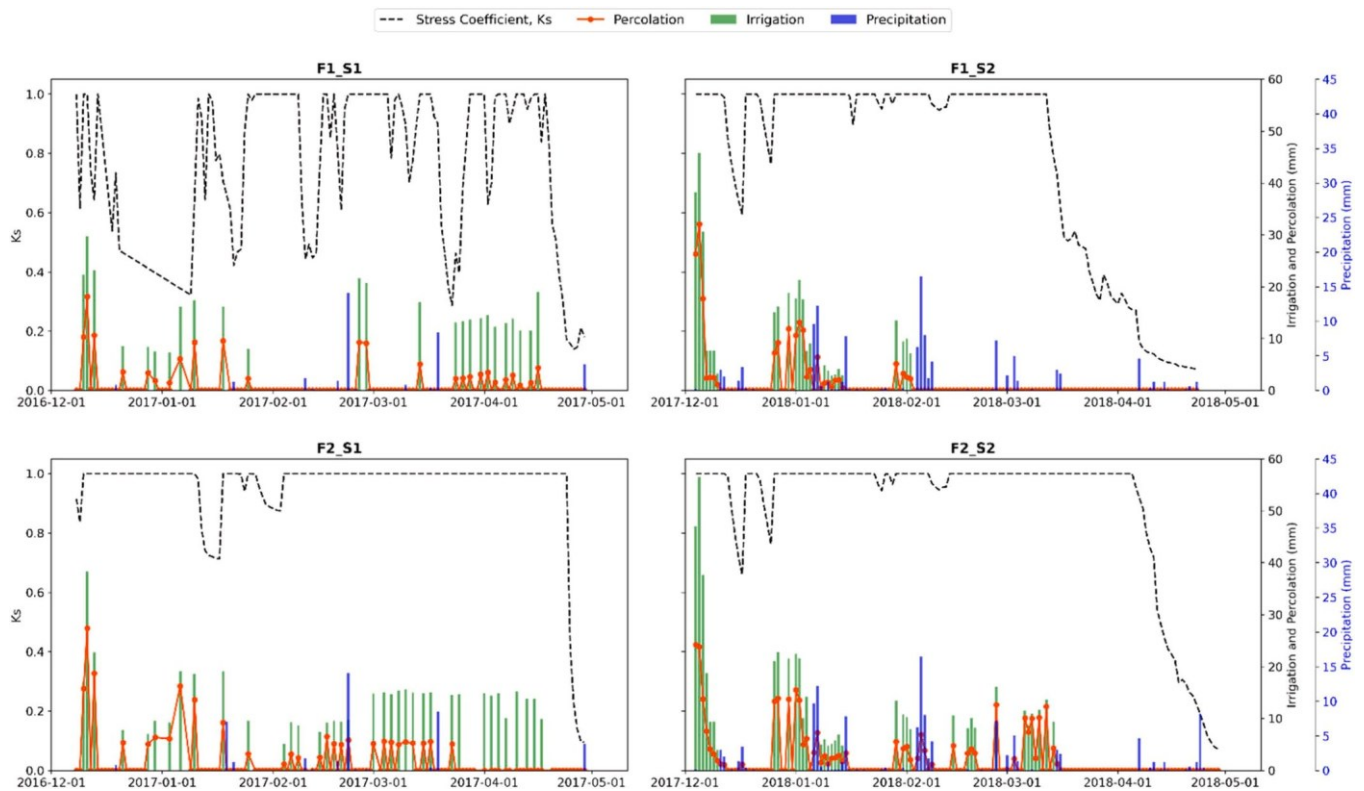


Fig. 6. Temporal variation of stress coefficient (K_s), irrigation, precipitation, and deep percolation during wheat seasons (2016/2017 and 2017/2018) in fields F1 and F2 under real irrigation conditions.

between what the model can simulate and how water is applied in the field. Optimization therefore becomes essential, not to correct model

performance, but to translate simulation results into more efficient irrigation decisions. By integrating optimization algorithms, irrigation

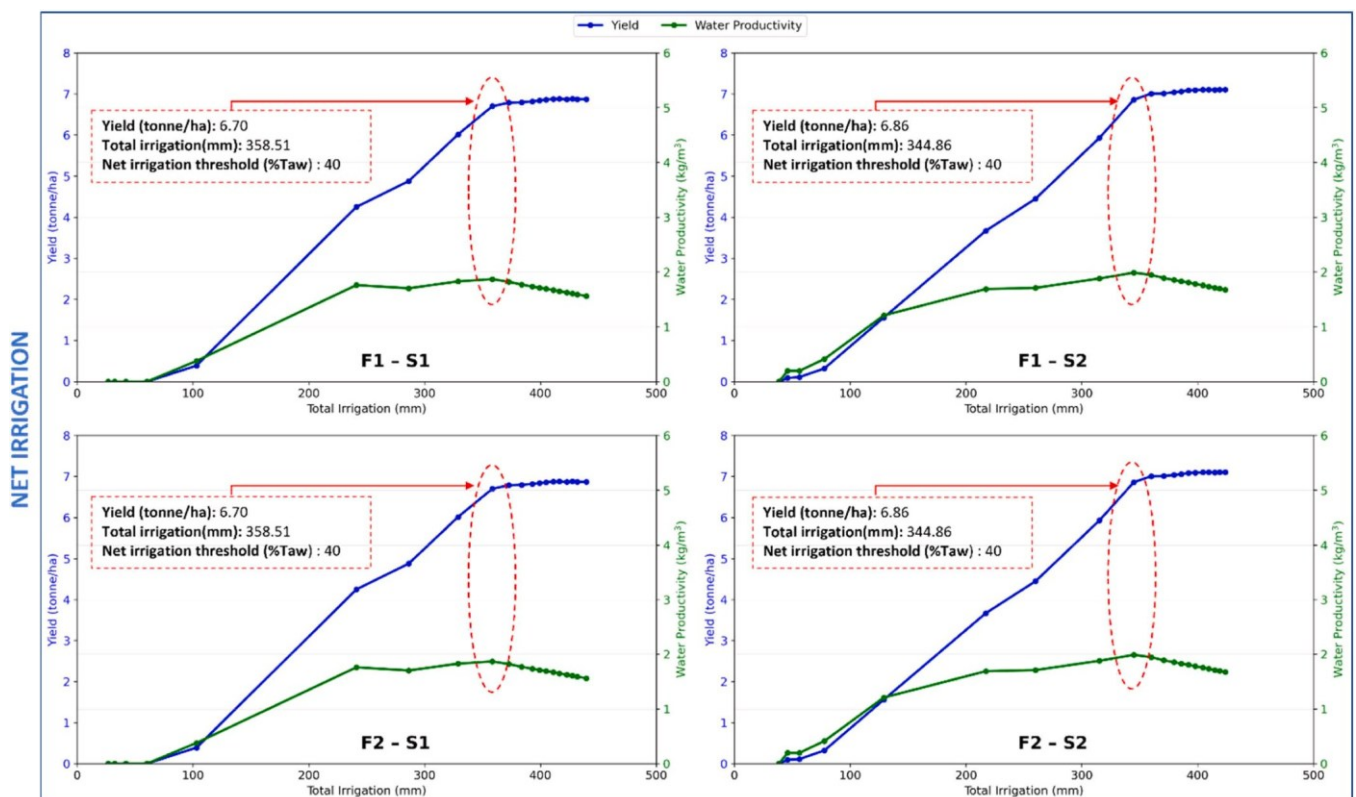


Fig. 7. Yield and water productivity response to net irrigation threshold.

thresholds can be dynamically adjusted to reduce losses, enhance yield, and improve WP under water-limited conditions.

3.3. Optimization of irrigation thresholds for improved water productivity and yield

This section presents the results of optimizing irrigation thresholds to maximize WP and Y in fields F1 and F2, using two distinct irrigation strategies: NET Irrigation and SMT Irrigation. Optimizing irrigation thresholds is crucial for improving water use efficiency by adjusting the amount of water applied to real crop needs. In this study, thresholds were selected to achieve an optimal balance between water use and productivity, considering the specificities of each field and the climatic conditions of both seasons. Figs. 7 and 8 display, for each tested threshold, the corresponding total seasonal irrigation, yield, and WP, which makes the optimization behaviour explicit: WP increases with irrigation up to an optimum and then declines once additional water no longer produces meaningful yield gains. The highlighted point in each panel marks the selected optimal threshold.

For the NET irrigation strategy, the optimization process identified a fixed threshold of 40% of (TAW) as optimal across fields and seasons. This value provided the best compromise between minimizing water use and maintaining high yields, avoiding excessive water stress while limiting deep percolation losses (Fig. 7). When the threshold was set below 40% TAW, the slight gains in Y were offset by a disproportionate increase in seasonal irrigation, reducing WP. Conversely, thresholds above 40% TAW led to more frequent and intense water stress episodes, which decreased Y and consequently WP. The optimized NET threshold of 40% TAW therefore represents a robust solution that stabilizes WP while preserving satisfactory Y under both S1 and S2 conditions. For the SMT Irrigation strategy, the optimization yielded stage-specific thresholds that varied between seasons. In S1, the optimal configuration corresponded to a relatively uniform threshold of 45% TAW applied during the development and mid-season stages, when water demand is highest

and critical for canopy expansion and grain formation. In S2, the optimal thresholds ranged from 30% to 60% TAW, with stricter depletion (30% TAW) during the early development stage and a more relaxed threshold (60% TAW) during mid-season (Fig. 8). These configurations outperformed alternative combinations because they better matched the crop's sensitivity at each phenological stage: lower depletion levels were sufficient during less demanding or better-supported periods, while higher thresholds prevented yield losses during critical phases of maximum canopy development. Overall, these optimized SMT patterns achieved higher WP while maintaining Y comparable to, or slightly higher than, those obtained under the NET strategy.

The contrast between the optimal SMT thresholds in S1 ([0, 45, 45, 0] % TAW) and S2 ([0, 30, 60, 0] % TAW) is consistent with the inter-annual variability in climatic conditions (Fig. 2). S2 received almost twice as much rainfall as S1, with several effective events distributed during the growing season, while ET₀ followed a broadly similar seasonal pattern in both years. In this wetter and more variable context, the optimization favoured a stricter threshold during the development stage (30% TAW) to prevent early water stress when rainfall is still limited and ET₀ begins to increase, and a more relaxed threshold during mid-season (60% TAW), where additional rainfall events and a more favourable precipitation–ET₀ balance allow larger soil water depletion without jeopardizing Y. Conversely, the drier S1 season, characterized by lower and less compensating rainfall, led to a more uniform optimal threshold of 45% TAW during the main growth stages, reflecting the need to maintain a more conservative soil water status throughout the cycle. For the NET strategy, the similar optimal threshold across seasons (40% TAW) reflects a compromise between these contrasting climatic regimes, providing sufficient protection against stress in the drier S1 while remaining efficient under the wetter S2 conditions.

These optimized thresholds tailored to each field's soil and climatic context and proved effective in reducing total irrigation volumes without compromising yield, resulting in higher WP. Building on these optimized thresholds, the next section quantifies their practical impact

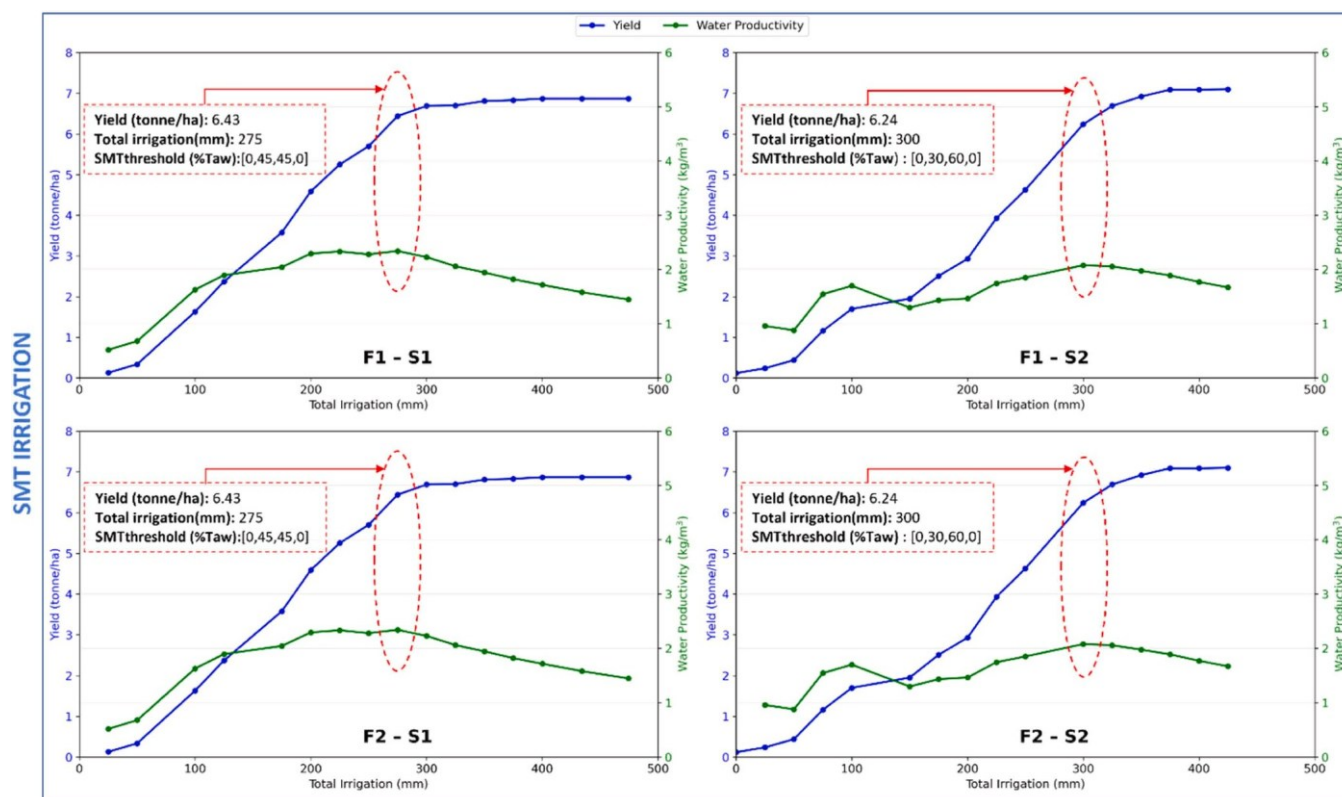


Fig. 8. Yield and water productivity response to growth stage-based soil moisture thresholds.

by comparing, for each field and season, irrigation volumes, ETa, yield, and WP before and after optimization under the Real, NET, and SMT strategies.

3.4. Impact of irrigation optimization on water use strategies: before and after optimization

Optimizing irrigation thresholds had a substantial impact on both water savings and productivity, enabling reductions in irrigation volumes while maintaining or enhancing crop yields. Figs. 9 and 10 summarize these improvements, showing how SMT consistently outperformed the Real and NET strategies in terms of irrigation volume and WP across both fields and seasons.

In F1-S1, irrigation volumes were higher before optimization, with 374.20 mm (Real), 400 mm (NET), and 400 mm (SMT), yielding WP values of 0.92, 1.71, and 1.72, respectively (Fig. 9). After implementing optimization, total irrigation quantity under SMT decreased from 400 mm to 275 mm, representing a 31.3% reduction, while WP increased from 1.72 to 2.33 kg/m³ (+35.5%). This reduction in irrigation highlights the positive effects of optimized irrigation thresholds, enabling more efficient water use while maintaining similar or even improved yields. In F2-S1, similar trends were observed (Fig. 10). Before optimization, irrigation volumes were 504.02 mm (Real), 400 mm (NET), and 400 mm (SMT), with corresponding WP values of 1.26, 1.71, and 1.71 kg/m³, respectively. After optimization, SMT irrigation was reduced to 275 mm (−31.3%), while WP increased from 1.71 to 2.33 kg/m³ (+36.3%), reinforcing the effectiveness of growth-stage-specific irrigation thresholds. In Season 2, optimization also reduced irrigation volumes: in F1-S2 and F2-S2, SMT irrigation decreased from 388.15 mm to 300 mm (−22.7%), with WP improving from 1.82 to 2.08 kg/m³ (+14.3%).

Across both fields, the optimized SMT strategy consistently outperformed the Real and NET approaches in terms of WP. These improvements are attributed to precise timing and volume adjustment based on crop phenology, which reduced unnecessary irrigation events and enhanced the match between water supply and crop demand. This analysis confirms that irrigation optimization, particularly via the SMT strategy, not only improves WP but also enhances sustainability in water-limited agricultural systems. By applying water more strategically, aligned with physiological crop stages and actual field conditions, it is possible to achieve high yields with reduced water input, making

these strategies especially valuable for semi-arid and drought-prone regions.

Complementing these graphical insights, Tables 4 and 5 provide a detailed breakdown of the water balance components, crop yield, and WP values (WP_{Irrigation} and WP_{ETa}) before and after optimization across both experimental fields. To better support management decisions, we also analysed the seasonal irrigation frequency and timing for each field, season, and strategy (Real, NET/NETopt, SMT/SMTopt). The number and distribution of irrigation events are summarised in the irrigation calendars presented in Appendix Figs. A.1–A.2 for F1-S1, F1-S2, F2-S1, and F2-S2. These calendars show that optimization, particularly SMTopt, not only reduced total irrigation volumes but also decreased the number of irrigation events and shifted them towards periods of higher crop water demand, thereby simplifying scheduling while improving WP. In addition to the seasonal water balance, we also analysed the temporal distribution of water stress by counting the number of days with K_s < 0.8 over the whole cycle and specifically during the flowering–grain filling period. Under the Real strategy, F1 was exposed to prolonged stress, with 48 and 87 days with K_s < 0.8 in S1 and S2, respectively, including 11 and 34 days during flowering–grain filling. F2 showed much fewer stress days (10 and 66 days in S1 and S2, with 0 and 12 days during flowering–grain filling), reflecting its initially over-irrigated management. After optimization, NET and SMT reshaped this stress pattern: NET reduced the number of stress days to 30 and 72 in S1 and S2 for both fields, and completely removed stress during flowering–grain filling (0 days with K_s < 0.8 in all cases). SMT maintained a similar total number of stress days (46 and 81 in S1 and S2), but strongly reduced stress during the reproductive stages compared with the Real strategy (0 and 11 days in S1 and S2 for both fields, respectively). This redistribution helps explain how SMT achieves higher WP_{Irrigation} by concentrating limited water resources on the reproductive phase, where yield is most sensitive to water deficits, while accepting moderate stress at less sensitive stages.

These tables provide the quantitative foundation for evaluating the performance of the three strategies across both seasons. While this section focused on seasonal water use and productivity indicators, the following analysis examines soil water content dynamics to show how optimization reshapes the temporal distribution of root-zone moisture, stress, and percolation under the NET and SMT strategies.

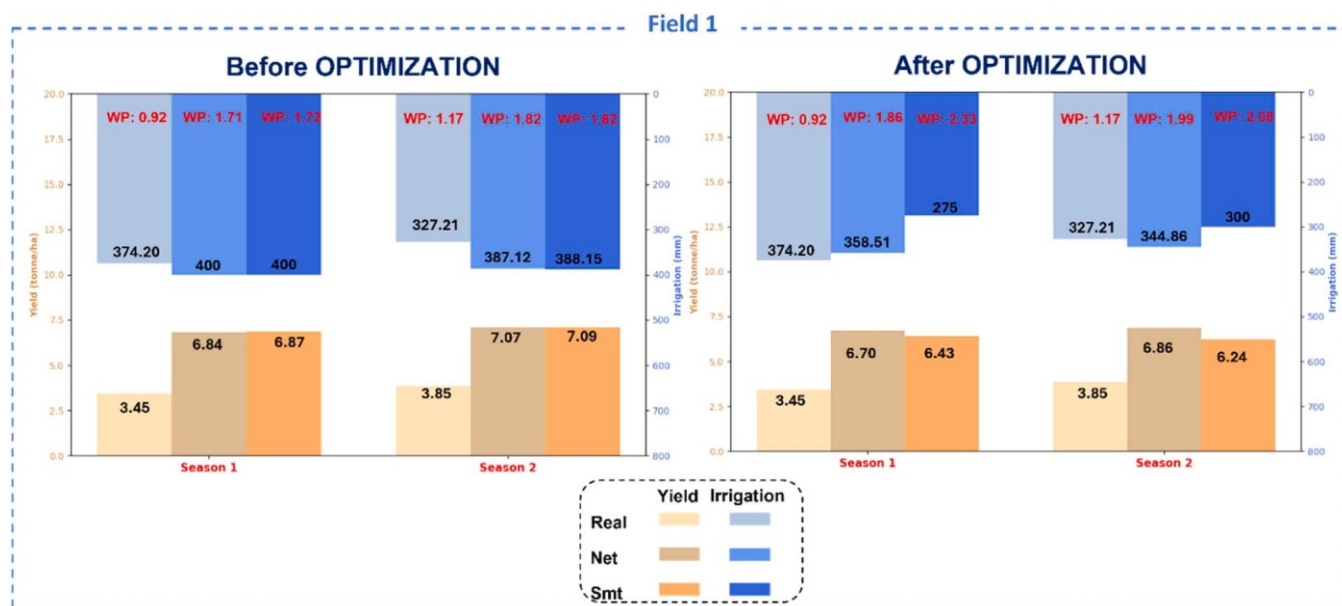


Fig. 9. Comparison of irrigation amount, evapotranspiration, yield, and water productivity under different strategies (Real, NET, SMT) before optimization.

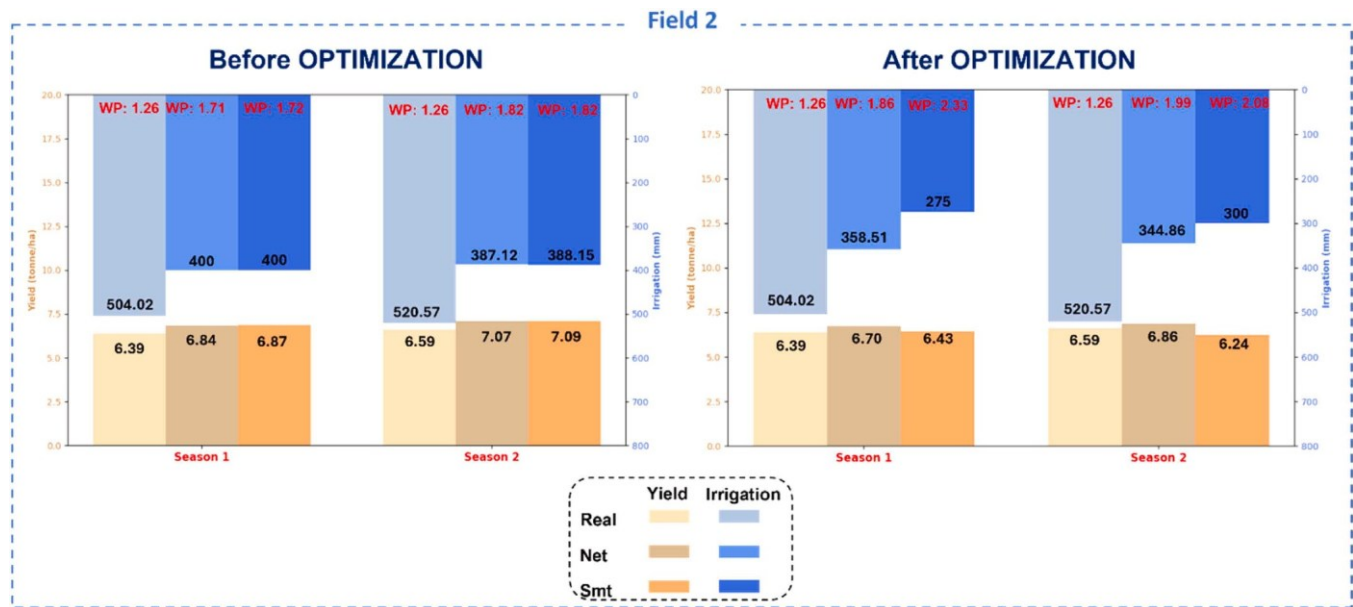


Fig. 10. Comparison of irrigation amount, evapotranspiration, yield, and water productivity under different strategies (Real, NET, SMT) after optimization.

Table 4

Comparative analysis of water balance for field F1 before and after irrigation optimization for two seasons: S1 and S2.

Water Balance				Irrigation (mm)	Precipitation (mm)	Eta (mm)	Percolation (mm)	Yield (t/ha)	WP (kg/m ³)	
									WP _{Irrigation}	WP _{ETa}
F1	Before	S1	Real	374.2	57.08	317	114.28	3.45	0.92	1.08
			NET	400	57.08	407.08	50	6.84	1.71	1.68
			SMT	400	57.08	407.08	50	6.87	1.72	1.68
		S2	Real	327.21	123.64	307.99	142.86	3.85	1.17	1.25
			NET	387.12	123.64	476.56	34.2	7.07	1.82	1.48
			SMT	388.15	123.64	476.86	34.93	7.09	1.82	1.47
	After	S1	NET	358.51	57.08	385.59	30	6.7	1.86	1.74
			SMT	275	57.08	302.08	30	6.43	2.33	2.13
		S2	NET	344.86	123.64	453.5	15	6.86	1.99	1.51
			SMT	300	123.64	407.64	16	6.24	2.08	1.53

Table 5

Comparative analysis of water balance for field F2 before and after irrigation optimization for two seasons: S1 and S2.

Water Balance				Irrigation (mm)	Precipitation (mm)	Eta (mm)	Percolation (mm)	Yield (t/ha)	WP (kg/m ³)	
									WP _{Irrigation}	WP _{ETa}
F2	Before	S1	Real	504.026	57.08	396.06	165	6.39	1.26	1.61
			NET	400	57.08	411.08	46	6.84	1.71	1.66
			SMT	400	57.08	411.08	46	6.87	1.72	1.67
		S2	Real	520.57	123.64	402.9	241	6.59	1.26	1.64
			NET	387.12	123.64	472.7	38	7.07	1.82	1.50
			SMT	388.15	123.64	472.82	39	7.09	1.82	1.50
	After	S1	NET	358.51	57.08	385.59	30	6.7	1.86	1.74
			SMT	275	57.08	302.08	30	6.43	2.33	2.13
		S2	NET	344.86	123.64	453.5	15	6.86	1.99	1.51
			SMT	300	123.64	407.64	16	6.24	2.08	1.53

3.5. Impact of irrigation optimization on soil water content dynamics and crop yield

Soil water content (WC) dynamics are critical to optimizing crop performance and achieving efficient water resource management. Figs. A.1- A.4 illustrate the relationship between WC, irrigation, and precipitation across both growing seasons, before and after optimization, under NET and SMT irrigation strategies in Fields F1 and F2. These visualizations highlight how optimized irrigation practices help maintain soil moisture within the ideal range while reducing overall water

use.

For the NET strategy, WC remained relatively stable due to consistent irrigation triggered by a fixed threshold of the (TAW). Before optimization (Figs. A.3- A.4), WC frequently hovered near or above Field Capacity (FC), indicating episodes of over-irrigation.

After optimization, irrigation volumes were reduced (e.g., from 400 mm to 358.51 mm in F1-S1), while WC was still maintained close to FC. This demonstrates the effectiveness of NET Irrigation in ensuring water availability for crops while simultaneously reducing excess water use. In contrast, SMT strategy exhibited more dynamic WC responses

due to its adaptive approach, with different thresholds applied at various crop growth stages. As seen in Figs. A.3 (F1) and A.4 (F2), higher thresholds were applied during critical stages like canopy development and peak canopy, maintaining WC near FC. During less demanding stages (e.g., emergence and senescence), lower thresholds allowed gradual reductions in WC. After optimization, SMT irrigation volumes decreased sharply (e.g., from 400 mm to 275 mm in F1, S1), yet WC remained within optimal bounds, proving SMT's adaptability in managing soil moisture efficiently.

Precipitation, highlighted in all figures, also played a significant role in supplementing irrigation, particularly during the second season for both fields. For example, in F1-S2, precipitation increased from 57.08 mm to 123.64 mm, reducing the reliance on irrigation to maintain adequate WC within desirable levels. The figures clearly show how spikes in precipitation temporarily boosted WC, resulting in fewer irrigation events. This underscores the importance of integrating natural rainfall into irrigation planning to enhance water-use efficiency. Monitoring WC is essential to assess soil water retention capacity and its availability for crops. Before optimization, excessive irrigation often resulted in WC exceeding FC, as seen in Figs. A.3-A.6, which could lead to inefficiencies and waterlogging. After optimization, irrigation practices were better aligned with crop water requirements, maintaining WC within an optimal range between FC and the wilting point (WP), as shown in Figs. A.4-A.6. This not only improved water-use efficiency but also minimized water losses due to percolation and evaporation.

The results highlight the effectiveness of optimized irrigation strategies in maintaining WC at levels that support crop growth while significantly reducing water use. By combining adapted irrigation practices with precipitation, significant water savings were achieved without compromising yields. These results emphasize the importance of WC monitoring and irrigation optimization for sustainable water and soil management in agriculture.

3.6. Influence of crop and soil parameters on simulated wheat yield across irrigation strategies

The OAT sensitivity analysis (Fig. 11) reveals a consistent hierarchy of influential parameters across irrigation strategies. In both the stressed (F1-S1) and well-irrigated (F2-S2) conditions, WP* and HI₀ were the most influential variables, with yield sensitivities around 18–21% for all three irrigation strategies. This confirms that the calibrated values of WP* and HI₀ are critical for reproducing wheat yield and that their dominance is robust to changes in irrigation management.

Under Real (farmer) irrigation in the stressed field (F1-S1), Z_{max} and θ_{FC} also showed marked influence on yield (sensitivities of about 15–21%), reflecting the strong control of rooting depth and soil water storage on crop performance when irrigation is poorly scheduled. Once irrigation thresholds were optimized (NETopt and SMTopt), the

sensitivity of yield to Z_{max} and θ_{FC} decreased markedly, whereas WP* and HI₀ remained dominant. This indicates that optimized irrigation partially compensates for uncertainty in soil hydraulic properties by maintaining more suitable soil-water conditions.

In the well-irrigated field (F2-S2), a similar pattern was observed. WP* and HI₀ again showed the highest and most stable sensitivities, while θ_{FC} and θ_{WP} were particularly influential under the Real scenario but became less critical under NETopt and SMTopt. CGC exhibited moderate sensitivity (≈ 10 –13%) in both fields, especially under NETopt and SMTopt, highlighting the role of canopy development in mediating the benefits of optimized irrigation.

The stability of the parameter ranking between NETopt and SMTopt, and across stressed (F1-S1) and well-irrigated (F2-S2) conditions, suggests that the calibrated crop parameters mainly reflect general wheat physiological behavior rather than site-specific artefacts. This supports the use of the calibrated parameter set for scenario analysis and for applying the optimized thresholds to other fields with similar cultivar and soil conditions in the Haouz Plain.

4. Discussion

This study evaluated how optimizing irrigation thresholds affects irrigation water use, evapotranspiration, yield, and WP in two wheat fields, F1 (stressed) and F2 (well irrigated), over two seasons. The results show that the proposed NET and SMT strategies can substantially reduce irrigation volumes and deep percolation while maintaining, and in some cases improving, yield and WP. In practical terms, part of the water that was previously lost as non-productive fluxes is converted into productive transpiration, increasing the amount of grain harvested per cubic metre of water applied.

Under farmer practices (Real), the combined analysis of Ks and soil-water dynamics revealed clear management shortcomings in both fields. In F1, Ks frequently dropped below 1, and often below 0.8, during sensitive growth stages, indicating repeated water stress at times when crop water demand was highest. In F2, irrigation was often applied close to or after rainfall events, and soil water content remained near or above field capacity for extended periods, pointing to over-irrigation and substantial percolation losses. These patterns reflect the absence of decision-support tools and the reliance on visual inspection and canal availability, rather than on soil water status and ET₀. As a result, rainfall was not efficiently integrated into the irrigation schedule: part of the seasonal precipitation simply added to an already wet profile, increasing drainage instead of reducing irrigation demand.

After optimization, both irrigation volumes and ET_a decreased, while WP increased, particularly under SMT. In both fields, SMT achieved large irrigation savings (on the order of 25–35% compared to pre-optimized SMT and farmer practices) with only minor changes in yield. WP_{irrigation} and WP_{ETa} both increased, confirming that water use became

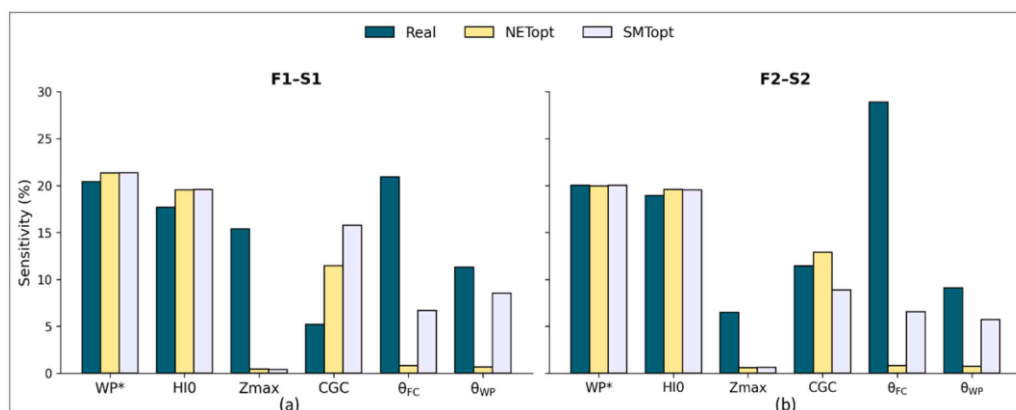


Fig. 11. Sensitivity of simulated wheat yield to key parameters under different irrigation strategies for (a) F1-S1 and (b) F2-S2.

more efficient rather than simply reduced. The irrigation and percolation components of the water balance show that SMT not only cuts back on the total irrigation applied but also reduces deep percolation relative to Real and pre-optimized SMT, indicating that more of the applied water is retained within the root zone and used for transpiration. This behaviour is particularly evident in the wetter second season, where SMT allows the soil to draw down more between rainfall events before triggering irrigation, so that precipitation contributes directly to replenishing TAW instead of exacerbating drainage.

Yield responses to optimization were generally small and straightforward to interpret. In the stressed field F1, SMT transformed a clearly sub-optimal situation characterized by poorly timed irrigation and strong yield penalties into a productive system: yield roughly doubled compared to Real while irrigation dropped markedly. In F2, which was already close to potential yield under full irrigation, the different strategies produced yields within a narrow range (on the order of $\pm 0.20 \text{ t ha}^{-1}$), despite substantial reductions in irrigation. These results support the idea that, in well-managed systems, drip irrigation combined with optimized thresholds can significantly improve WP without compromising yield, whereas in poorly managed systems it can both recover yield and reduce water losses by correcting the timing of applications.

The comparison between NET and SMT can be better understood when linked to wheat physiology and soil hydrology. Wheat demand and sensitivity to water stress are highest during canopy expansion, jointing, heading, and grain filling, and lower during emergence and late senescence. The NET strategy uses a single depletion threshold throughout the cycle. This stabilizes Ks and protects the crop against severe stress, but it also tends to trigger irrigation even in stages where the crop can tolerate greater soil water depletion. As a result, NET sometimes refills the soil profile when it is still relatively wet or shortly before rainfall, favouring percolation losses and limiting the effective contribution of rainfall to root-zone replenishment. In practice, this typically translates into more frequent irrigation events distributed fairly evenly across the season, including periods when the crop is physiologically less sensitive to stress.

In contrast, SMT uses stage-specific thresholds aligned with crop sensitivity. Higher thresholds are applied during canopy development and grain filling, keeping Ks close to 1 during these critical stages and protecting yield, while lower thresholds during emergence and senescence allow greater depletion without affecting final production. When combined with rainfall, this dynamic scheme allows the profile to dry more before irrigation is triggered, especially in S2, so that rainfall is more fully used and percolation reduced. The analysis of Ks confirms this behaviour: under Real, F1 experienced several dozen days with $K_s < 0.8$, including numerous days during flowering–grain filling, whereas under SMT the number of stress days in that window was largely eliminated or sharply reduced. SMT therefore outperforms NET not only because it saves more water, but because it reshapes the temporal distribution of stress to minimise deficits during the most sensitive phases while accepting moderate stress when the crop can tolerate it and concentrates irrigation events during those key stages rather than spreading them uniformly.

The slight yield reductions observed under SMT in a few cases, such as F1–S1 when compared to the pre-optimized SMT baseline, illustrate the classical trade-off between maximum yield per hectare and maximum yield per cubic metre of water. In these situations, substantial reductions in irrigation (on the order of 100–150 mm) were associated with relatively small decreases in yield (on the order of a few tenths of a tonne per hectare). Under severe water scarcity, such trade-offs are often acceptable or even desirable, because the saved water can be reallocated to other fields or crops. For decision-makers, the key question becomes whether the priority is to maximise absolute yield or to maximise WP, the present results provide quantitative examples showing that modest yield sacrifices can deliver large water savings and significant gains in WP.

The optimized NET threshold can also be interpreted in the context

of previous work. [Toumi et al. \(2016\)](#) proposed a depletion fraction $Dr = 0.6$ as an effective management rule for wheat in semi-arid Morocco, corresponding to irrigation when 40% of TAW remains. In this study, this recommendation was used as a baseline in AquaCrop-OSPy and then refined by the optimizer under local conditions. The optimisation converged towards an optimal NET threshold of around 40% TAW, which lies within the same conceptual range as the $Dr = 0.6$ rule but is adapted to drip irrigation, the specific climate of the study years, and an objective function that explicitly maximizes WP rather than yield alone. This convergence suggests that the model-based optimization framework is consistent with earlier empirical guidelines while providing a more nuanced, site-specific adjustment of depletion thresholds.

The OAT sensitivity analysis further supports the robustness and generalizability of the calibrated AquaCrop-OSPy parameters. Perturbing key crop parameters (WP^* , HI_0 , CGC, Zr_{max}) and soil hydraulic properties (θ_{FC} , θ_{WP}) by $\pm 20\%$ showed that yield is most sensitive to WP^* , and HI_0 while Zr_{max} and soil properties exert a moderate but non-negligible influence. Importantly, the ranking of influential parameters remained similar across irrigation regimes (Real, NETopt, SMTopt) and across the stressed and well-irrigated fields. This stability indicates that the calibration primarily captured general physiological behaviour of winter wheat under the local conditions, rather than overfitting to a single irrigation scenario, and provides additional confidence in using the calibrated parameter set for scenario analysis and transfer to similar fields in the Haouz Plain.

These findings are consistent with previous research on model-based irrigation management in semi-arid systems. ([Bouswir et al., 2025](#)) demonstrated that optimised irrigation schedules using SIMDualKc can strongly reduce deep percolation while avoiding water stress in winter wheat, and ([Toumi et al., 2016](#)) and ([Hadria et al., 2007](#)) showed that crop models can help achieve both water savings and yield stability in Moroccan conditions. More recently, ([Yang et al., 2023](#)) reported that drip irrigation improved yield and crop WP in spring wheat by enhancing crop development and water-use efficiency. Our results confirm these mechanisms in a different modelling framework and go further by explicitly coupling AquaCrop-OSPy with gradient-based optimization, evaluating both $WP_{Irrigation}$ and WP_{ETa} and testing the strategies under real farmer-managed stress conditions. Compared with ([Akbari et al., 2024](#)), who used AquaCrop with PSO for parameter calibration in saffron without optimising irrigation thresholds, and ([Jin et al., 2020](#)), who assimilated remote-sensing biomass and CC into AquaCrop for maize yield estimation, the present study focuses directly on threshold-based irrigation scheduling. In relation to ([Sharafkhane et al., 2024](#)), who developed an AquaCrop plug-in–PSO framework for forecast-based maize irrigation, our work provides an operational, field-tested evaluation of optimised NET and SMT thresholds for drip-irrigated wheat, including explicit before/after comparisons of irrigation volumes, percolation, ETa, and WP across two contrasting fields. These elements together highlight the specific methodological and applied contribution of this study.

Several limitations must nevertheless be acknowledged. As reported in previous evaluations ([Farahani et al., 2009](#); [Hussein et al., 2011](#); [Iqbal et al., 2014](#); [Mkhabela and Bullock, 2012](#); [Toumi et al., 2016](#)), AquaCrop tends to perform better for canopy cover, ETa, and yield than for TWC, often showing seasonal overestimation. The present simulations confirm this behaviour, with TWC tending to remain slightly higher than observations, especially during wet periods. This bias is likely linked to simplified assumptions in the soil–water balance, including the representation of deep percolation and subsoil processes, and contributes to uncertainty in the absolute magnitude of percolation and storage changes. For this reason, calibration and validation in this study focused on canopy cover, biomass, and ETa, which provide more reliable indicators of crop response. While this limitation does not undermine the relative comparison between irrigation strategies, it suggests that absolute TWC and percolation values should be interpreted with caution, particularly when extrapolating to groundwater recharge or basin-scale

hydrology. In addition, AquaCrop does not explicitly represent nutrient dynamics or detailed post-stress recovery mechanisms, which may affect the realism of long stress–recovery cycles and the response to combined water and nutrient deficits. A further limitation is that formal statistical significance tests were not systematically applied to all differences in WP and yield; interpretation therefore relies on the magnitude and consistency of the simulated changes, and future work should include a more comprehensive statistical assessment.

Finally, the generalizability of the optimised thresholds warrants careful consideration. The specific NET and SMT values identified here are tailored to the climate, soil, irrigation system, and cultivar of the Chichaoua site. Extending them to other areas of the Haouz Plain or to different semi-arid regions will require local calibration of crop and soil parameters, and possibly adjustment of depletion thresholds to account for differences in salinity, rooting depth, soil texture, and groundwater depth. At farm and regional scales, additional challenges include the availability of field data, the computational cost of running many model–optimization combinations, and the need to integrate farmer constraints such as labour, canal rotation schedules, and energy costs. Future work should therefore focus on combining AquaCrop-OSPy and optimisation with real-time soil moisture sensing, remote-sensing indicators of canopy status, and data assimilation or machine-learning approaches to update thresholds dynamically during the season. Larger multi-field and multi-farm experiments in the Haouz Plain and comparable semi-arid environments would also help to assess the robustness of NETopt and SMTopt under a wider range of soils, cultivars, and management practices. These developments would strengthen the use of AquaCrop-OSPy-based optimisation as an operational decision-support tool for sustainable irrigation under increasing water scarcity.

5. Conclusion

This study demonstrates that coupling the AquaCrop-OSPy model with gradient-based optimization algorithms provides an effective decision-support framework for designing dynamic irrigation schedules in drip-irrigated wheat systems under semi-arid conditions. By treating soil water depletion thresholds as decision variables and optimizing them with respect to irrigation water productivity, the proposed framework generates irrigation schedules that substantially reduce non-productive water use while maintaining agronomically acceptable yields. Among the strategies tested, the Soil Moisture Target approach proved more effective than both farmers' practices and the fixed-threshold NET strategy, as it better aligns root-zone soil moisture with crop phenology and the crop's sensitivity to water stress.

From a scientific and methodological perspective, the study formalizes the use of AquaCrop-OSPy as a “black-box” simulator embedded within a gradient-based optimization scheme. In this framework, the minimize_scalar algorithm is used to optimize a single depletion threshold for the NET strategy, while the L-BFGS-B algorithm is applied to optimize four phenology-specific depletion thresholds for the SMT strategy. This combination illustrates how a process-based crop model can be integrated into optimization routines to explore the trade-offs between water savings and yield preservation. The results highlight the importance of stage-specific control of soil water depletion, showing that maintaining higher soil moisture during canopy development and grain filling, while allowing greater depletion during early and late stages, can substantially improve irrigation water productivity compared with a single, season-long threshold.

Practically, the findings support the implementation of SMT-based irrigation management for drip-irrigated wheat in the Haouz Plain and similar semi-arid regions. For farmers and irrigation managers, this translates into applying higher soil-moisture thresholds during rapid canopy development and grain filling, and lower thresholds during emergence and late-season stages, using either soil water depletion estimates or field sensor readings. The optimized thresholds derived in this study can be converted into operational rules and used to complement

existing drip infrastructure without major additional investment. This work provides actionable threshold solutions for wheat drip irrigation systems in Morocco's Haouz Plain, contributing to the alleviation of dual pressures from water scarcity and food security.

This work also has limitations. The analysis is based on two fields and two growing seasons in a single semi-arid context, and AquaCrop-OSPy simulations were run deterministically without experimental or stochastic replicates. Consequently, differences in yield, irrigation water use, and water productivity between scenarios are interpreted descriptively rather than supported by formal statistical significance tests. Future research should extend this approach to a broader range of crops, soils, and climatic conditions, integrate real-time soil moisture and remote sensing data to drive SMT-based scheduling in operational settings, and explore hybrid frameworks that combine process-based modelling, optimization, and machine learning to better quantify uncertainty and adapt thresholds under climate variability.

The results of this study demonstrate that integrating AquaCrop-OSPy with gradient-based optimization transforms the model into a practical tool for designing high-efficiency irrigation strategies in water-scarce environments. By providing a rigorous yet operational framework to optimize soil moisture thresholds for drip-irrigated wheat, this study directly addresses the challenge of producing more food with less water in semi-arid Morocco and in other regions facing intensifying water scarcity.

CRedit authorship contribution statement

Salah Er-Raki: Writing – review & editing, Supervision, Investigation, Conceptualization. **Abdelghani Chehbouni:** Writing – review & editing, Project administration. **El Houssaine Bouras:** Writing – review & editing, Supervision, Investigation. **Oumaima Kaissi:** Writing – original draft, Software, Methodology, Formal analysis. **Lamia Jallal:** Writing – review & editing. **Salwa Belqaziz:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.agwat.2026.110418](https://doi.org/10.1016/j.agwat.2026.110418).

Data availability

Data will be made available on request.

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